1. Competency #1: Perform a simple calculation

> *"Perform a simple calculation"*

- Highly relevant: The core of the task is a standard percentage change computation:

$$\frac{110-78}{78} \times 100\% = \frac{32}{78} \times 100\% \approx 41\%$$

- Although it involves division and multiplication, the steps are straightforward and require no complex modeling—this is basic arithmetic, fully aligned with Competency #1.

---

2. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process (e.g., choosing appropriate precision)

> *"Use an understanding of the context to guide or expedite the mathematical solving process..."*

- Highly relevant: The stem provides crucial contextual cues:
  - "About 78 grams" and "about 110 grams" indicate the data are approximate.
  - Therefore, the answer need not be exact to several decimal places; a reasonable estimate suffices.
- Students can quickly approximate: 32 ÷ 78 ≈ 0.41 → 41%, without a calculator. Even if they compute 41.03%, they should round appropriately to match the given choices.
- The word "about" directly signals the use of suitable precision, preventing over-computation.

---

3. Competency #7: Manipulate numbers, graphical and statistical data, algebraic expressions, equations, and geometric representations

> *"Manipulate numbers, graphical and statistical data... algebraic expressions and equations..."*

- Strongly relevant: Students must:
  - Identify two numerical values (78 and 110);
  - Compute their difference (110 − 78);
  - Apply the percentage formula: (difference ÷ original) × 100%;
  - Carry out the arithmetic.
- Although no graphs or algebraic symbols are involved, the manipulation of numbers and formulas constitutes fundamental data handling, fitting squarely within Competency #7.

4. Competency #3: Implement a given strategy to determine a mathematical solution

> *"Implement a given strategy to determine a mathematical solution"*

- Moderately relevant: While the question doesn't explicitly state "use the percentage change formula," this method is a standard, curriculum-taught strategy that students are expected to recall and apply.
- Thus, it qualifies as implementing a known procedure, aligning with Competency #3.

---

5. Competency #5: Understand and utilise constructs based on definitions, rules and formal systems, including employing familiar algorithms

> *"Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms"*

- Moderately relevant: Percentage increase is a clearly defined mathematical concept governed by a fixed rule:

$$\frac{\text{new} - \text{original}}{\text{original}} \times 100\%$$

- Students must invoke this standardized algorithm, demonstrating understanding and application of a formal mathematical structure.

1. 第1项：进行简单计算

> *"Perform a simple calculation"*

- 高度相关：本题的核心是执行一个标准的百分比变化计算：

$$\frac{110-78}{78} \times 100\% = \frac{32}{78} \times 100\% \approx 41\%$$

- 尽管涉及除法和乘法，但步骤直接、无复杂建模，属于基础算术操作，完全契合第1项。

---

2. 第13项：利用对情境的理解来指导或加速数学求解过程（例如采用符合情境的精度）

> *"Use an understanding of the context to guide or expedite the mathematical solving process..."*

- 高度相关：题干提供关键背景：
  - “第一枚约78克，第二枚约110克”中的“约”表明数据为近似值；
  - 因此，答案无需精确到小数点后多位，只需合理估算即可。
- 学生可快速判断：32 ÷ 78 ≈ 0.41 → 41%，而无需计算器；若算出41.03%，也应四舍五入匹配选项。
- 情境中的“约”字直接提示使用适当精度，避免过度计算。

---

3. 第7项：操作数字、图形与统计信息、代数表达式、方程及几何表示

> *"Manipulate numbers, graphical and statistical data... algebraic expressions and equations..."*

- 强相关：学生需：
  - 识别两个数值（78 和 110）；
  - 构造差值（110 – 78）；
  - 建立百分比公式（差值 ÷ 原始值 × 100%）；
  - 执行数值运算。
- 虽未涉及图表或代数符号，但数值与公式的操作是典型的数据处理行为，属于第7项范畴。

4. 第3项：实施给定策略以确定数学解

> *"Implement a given strategy to determine a mathematical solution"*

- 中等相关：虽然题目未明确写出“使用百分比变化公式”，但该公式是课程中标准策略，学生被期望调用这一熟悉方法。
- 因此，这属于“实施已知策略”（而非自主发明新方法），符合第3项精神。

---

5. 第5项：理解并运用基于定义、规则和形式系统的结构，包括使用熟悉算法

> *"Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms"*

- 中等相关：百分比变化是一个明确定义的数学概念，其计算遵循固定规则（(新−旧)/旧 × 100%）。
- 学生需调用这一标准化算法，体现对形式规则的理解与应用。